

EX:02**Design and Implement LLM-Based Applications for Question Answering, Text Summarization, and Sentiment Analysis****Aim:**

To design and implement Large Language Model (LLM)-based applications for Question Answering, Text Summarization, and Sentiment Analysis using the Google Gemini API, and evaluate their performance on real-world textual data.

Algorithm:

1. Install the required Python libraries (google-genai).
2. Obtain a Google Gemini API key and initialize client authentication.
3. Initialize the Gemini client and specify the model (gemini-3.5-flash-lite).
4. Define textual inputs, queries, and ground-truth benchmarks for Question Answering, Text Summarization, and Sentiment Analysis.
5. Construct a Question Answering prompt using context and query, invoke the model, and calculate the QA Score based on keyword overlap with expected answers.
6. Construct a Text Summarization prompt, generate a concise summary, and compute the Summary Similarity Score using unique vocabulary token overlap against the reference summary.
7. Construct a sentiment evaluation dataset with labeled samples (Positive, Negative, Neutral), iteratively prompt the model for strict label classification, and calculate Sentiment Accuracy.
8. Execute the applications using the Google GenAI SDK.
9. Record and display the generated outputs alongside task-specific evaluation metrics.
10. Aggregate and display the final evaluation metrics (QA Score, Summarization Similarity, and Sentiment Accuracy) to assess overall LLM performance.

Code:

```
!pip install -q google-genai

import os
from google import genai

os.environ["GEMINI_API_KEY"] = "aidjbahdbahbdkabdkjsabd-dasjsd--adnjsan"

client = genai.Client(api_key=os.environ["GEMINI_API_KEY"])
model = "gemini-3.5-flash-lite"

qa_text = """
Artificial Intelligence is a branch of computer science that focuses on creating
systems capable of performing tasks that normally require human intelligence.
These tasks include learning, reasoning, problem solving, understanding language,
and recognizing patterns.
"""

qa_question = "What are the main tasks performed by Artificial Intelligence?"

qa_expected = [
    "learning",
    "reasoning",
    "problem solving",
    "understanding language",
    "recognizing patterns"
]

summary_text = """
```

Artificial Intelligence has become an important technology in modern healthcare. It is used for disease detection, medical image analysis, drug discovery, patient monitoring, and personalized treatment. AI can analyze large amounts of medical data quickly and help doctors make better decisions. However, challenges such as data privacy, bias, and reliability must be carefully addressed.

```
summary_expected = """
AI is used in healthcare for disease detection, medical image analysis,
drug discovery, patient monitoring, and personalized treatment. It helps
analyze medical data and support doctors, but privacy, bias, and reliability
remain important challenges.
"""
```

```
sentiment_data = [
    ("I love this product. It works perfectly and I am very happy.", "Positive"),
    ("The product is terrible and completely useless.", "Negative"),
    ("The product arrived yesterday and has several features.", "Neutral"),
    ("The service was excellent and the staff were very helpful.", "Positive"),
    ("I am disappointed with the poor quality of this product.", "Negative")
]
```

```
qa_prompt = f"""
Answer the following question using the given text.
```

```
Text:
{qa_text}

Question:
{qa_question}
"""
```

```
summary_prompt = f"""
Summarize the following text in 2-3 sentences.
```

```
Text:
{summary_text}
"""
```

```
qa_response = client.models.generate_content(
    model=model,
    contents=qa_prompt
)
```

```
summary_response = client.models.generate_content(
    model=model,
    contents=summary_prompt
)
```

```
print("=" * 60)
print("QUESTION ANSWERING")
print("=" * 60)
print(qa_response.text)
```

```
qa_score = sum(
    word.lower() in qa_response.text.lower()
    for word in qa_expected
) / len(qa_expected) * 100
```

```
print("\nQA Score:", round(qa_score, 2), "%")
```

```
print("\n" + "=" * 60)
print("TEXT SUMMARIZATION")
print("=" * 60)
print(summary_response.text)
```

```
summary_words = set(summary_response.text.lower().split())
```

```

expected_words = set(summary_expected.lower().split())

summary_score = len(summary_words & expected_words) / len(expected_words) * 100

print("\nSummary Similarity Score:", round(summary_score, 2), "%")

print("\n" + "=" * 60)
print("SENTIMENT ANALYSIS")
print("=" * 60)

correct = 0

for text, expected in sentiment_data:

    prompt = f"""
Classify the sentiment of the following text as exactly one of:
Positive, Negative, Neutral.

Text:
{text}

Return only the sentiment label.
"""

    response = client.models.generate_content(
        model=model,
        contents=prompt
    )

    predicted = response.text.strip()

    if expected.lower() in predicted.lower():
        correct += 1

    print("\nText:", text)
    print("Expected:", expected)
    print("Predicted:", predicted)

sentiment_accuracy = correct / len(sentiment_data) * 100

print("\nSentiment Accuracy:", round(sentiment_accuracy, 2), "%")

print("\n" + "=" * 60)
print("FINAL EVALUATION")
print("=" * 60)

print("Question Answering Score:", round(qa_score, 2), "%")
print("Summarization Similarity:", round(summary_score, 2), "%")
print("Sentiment Accuracy:", round(sentiment_accuracy, 2), "%")

```

Output:

```

=====
QUESTION ANSWERING
=====
Based on the provided text, the main tasks performed by Artificial Intelligence are:

* Learning
* Reasoning
* Problem solving
* Understanding language
* Recognizing patterns

QA Score: 100.0 %

=====
TEXT SUMMARIZATION
=====

```

Artificial Intelligence has become a vital tool in modern healthcare, assisting doctors with disease detection, data analysis, and personalized treatments. However, its widespread adoption comes with important challenges that must be addressed, including data privacy, algorithmic bias, and system reliability.

Summary Similarity Score: 32.26 %

=====

SENTIMENT ANALYSIS

=====

Text: I love this product. It works perfectly and I am very happy.
Expected: Positive
Predicted: Positive

Text: The product is terrible and completely useless.
Expected: Negative
Predicted: Negative

Text: The product arrived yesterday and has several features.
Expected: Neutral
Predicted: Neutral

Text: The service was excellent and the staff were very helpful.
Expected: Positive
Predicted: Positive

Text: I am disappointed with the poor quality of this product.
Expected: Negative
Predicted: Negative

Sentiment Accuracy: 100.0 %

=====

FINAL EVALUATION

=====

Question Answering Score: 100.0 %
Summarization Similarity: 32.26 %
Sentiment Accuracy: 100.0 %

Result:

Thus, LLM-based applications for Question Answering, Text Summarization, and Sentiment Analysis were successfully designed, implemented, and evaluated using Google Gemini API.